

Reply to RFQ / BOM quotation — T-2SJ3W1113248A

Product: Racecar-35 Rev D integrated controller

Quantity: 5 units · turnkey PCB assembly (parts + assembly + bare PCB)

Date: 29 September 2026 · **Quote reference:** T-2SJ3W1113248A

Please apply the decisions below and issue the **revised quotation**. A corrected

BOM is attached (Racecar-RevD-BOM-pcbway-assembly-REV1.csv) — please re-match

from that file rather than the original.

1. Answers to the three "substitute OK?" questions

Item	Original	Your proposed substitute	Our answer
21	GRM21BR71H103KA01L (C10)	CL21B103KBANNNC	APPROVED — same value: 10 nF, 50 V, X7R, 0805
22	GRM21BR71H225KA01L (C11)	CL21A225KBQNNNE	NOT APPROVED — this is X5R (85 °C). The original is X7R (125 °C) and this part sits on the 3.3 V MCU rail. Please use CC0805KKX7R9BB225 (Yageo, LCSC C125847) or CL21B225KBYNNNE (Samsung, LCSC C2762602)
35	GRM21BR71H102KA01L (C18)	CC0805KRX7R9BB102	APPROVED — same value: 1 nF, 50 V, X7R, 0805

2. Answers to the four "out of stock" items

Item	Ref	Original	Our decision
1	U2	4091 / D36V50F5 (Pololu)	Remove from the assembly — we will fit this module ourselves. It is a socketed/headered raised module and needs no PCB change. Do not source it, do not charge assembly for it.
13	U1	TEENSY41 (PJRC Teensy 4.1)	Remove from the assembly — we will fit the Teensy ourselves (it goes into the supplied 2× 1×24 socket). Do not source it, do not charge assembly for it.
11	D2	SS14 (Diodes Inc)	Please use onsemi SS14 (LCSC C83852, stock >100 000). Alternative: Vishay SS14-E3/61T (LCSC C47460). Identical 40 V / 1 A SMA Schottky.
28	D5	PESD3V3U1UL, 315	Please use Nexperia PESD5V0F1BL, 315 (LCSC C45961, stock >50 000) — same

Item	Ref	Original	Our decision
			SOD-882 / DFN1006-2 land pattern. Alternative if unavailable: PESD5V0U1UL,315 (LCSC C85401).
48	F3 F4 F5	1206L010/30YR	This part number does not exist in the Littelfuse 1206L series (the 0.10 A device is only made as 1206L010/60, i.e. 1206L010/60WR). Please use 1206L010/60WR (LCSC C2153714, stock ~7 000; also Newark 10 764 / Digi-Key 4 861). Same 1206 package, same 100 mA hold current, 60 V instead of 30 V.

3. The other two notes

Item	Note	Our reply
14	BT1 1058 — "quoted part number is 1058, not CR2032"	1058 is correct. It is the Keystone SMD 20 mm coin-cell holder. "CR2032" in the description is the cell , which is not part of the BOM and which we fit by hand after assembly. Please supply Keystone 1058.
24	U8 NEO-M9N-00B — "price is changing higher"	Understood. Please confirm the final unit price with us before production starts. If it exceeds US\$20 per unit please stop and tell us before ordering.

4. One part-name warning

Item 2, TRACO TSR 1-2433: this must be **genuine TRACO Power**. A module with the same generic name is sold on LCSC (C53183919) but the brand is *YLPTEC* — that is **not acceptable** as the TRACO part. Please quote and supply genuine TRACO Power from an authorised distributor (Digi-Key, TME, Newark or Sager all have it in stock).

5. What is unchanged

- The **BOM structure, references, quantities and footprints** are unchanged — every substitution above is the same package/footprint, so ****no PCB change is required****.
- The **bare PCB** settings from your quote are accepted as-is: 4 layers, 150.05 × 155.05 mm, 1.6 mm, 1 oz outer copper, **ENIG**, impedance control ON (50 Ω single-ended on the GNSS feed), white silkscreen, green mask, do not panelise, keep the antenna notch in the profile.

- Substitutions above are only in the **BOM / parts list**.

6. What we need back from you

- The **revised quotation** for 5 units, built from the attached corrected BOM,

with U1 (Teensy 4.1) and U2 (Pololu) removed.

- Confirmation that all five corrected MPNs in §1 and §2 are ****matched and in**

stock in your system******, with their lead times.

- The final confirmed price for NEO-M9N-00B before production (see §3).
- Your ****PCB Assembly (turnkey) uploading instructions confirmed for the file**

Racecar-RevD-BOM-pcbway-assembly-REV1.csv****** — if your console requires the

project's original file name, we can rename it back on request.

Summary of part changes (for your BOM edit)

Ref	Out	In
C10	GRM21BR71H103KA01L (Murata)	CL21B103KBANNNC (Samsung)
C11	GRM21BR71H225KA01L (Murata)	CC0805KKX7R9BB225 (Yageo)
C18	GRM21BR71H102KA01L (Murata)	CC0805KRX7R9BB102 (Yageo)
D2	SS14 (Diodes Incorporated)	SS14 (onsemi)
D5	PESD3V3U1UL, 315 (Nexperia)	PESD5V0F1BL, 315 (Nexperia)
F3 F4 F5	1206L010/30YR (Littelfuse)	1206L010/60WR (Littelfuse)
U1 U2	Teensy 4.1, Pololu 4091	deleted from the BOM — customer supplied